Marketing Analytics: Assignment 02

An e-commerce company wants to understand what drives customers to convert (make a purchase) after being exposed to its marketing campaigns. The dataset marketing_conversion_data.csv contains data on 800 customers. Variables in the data set include:

- discount_pct: Discount percentage offered to the customer during the campaign (%)
- email_opens: Number of marketing emails opened by the customer in the last 30 days
- income: Customer's annual income, in $1,000s
- loyalty_yrs: Number of years the customer has been enrolled in the loyalty program
- site_mins: Average time spent on the company website per session, in minutes
- convert: Binary outcome variable (1 = converted/purchased, 0 = did not convert)

[1] Read the data set into your R environment as myData. Evaluate the structure of the myData data set. Do we have any missing value here? Explain.

[2] Estimate and interpret the linear probability model. What is the limitation of this model?

[3] Estimate the logistic regression model. Why do you think the logistic regression model is better at predicting a binary target variable? What is the difficulty of explaining the slope coefficient (partial effect) of this model?

[4] Explain the partial effect (slope coefficient) of each predictor variable on the odds of convert.

[5] Predict the conversion probability of the following two customers by using both models. What will be their binary value? Explain.

discount_pct	email_opens	income	loyalty_yrs	site_mins
10	3	55	1	6
28	9	45	2	14

[6] Calculate the accuracy of both the linear model and the logistic model.

[7] Use simulation to highlight the partial effect of emal_opens on the predicted convertion probability holding all other predictors at their mean values. What can you conclude from this simulation? Explain.

[8] Use the holdout method to compare the accuracy rates of two competing logistic regression models for convert, using the first 400 observations for training and the remaining 400 observations for validation. Model_1 uses all the predictors but Model_2 uses only the statistically significant predictors. Calculate the accuracy rates of both competing models.

[9] Does the selection of classification model (Model_1 vs Model_2) contradict the selection on the basis of AIC? Explain. Explain what model would you finally select? Why?

[10] Make a presentation on the basis of your analyses from 1 to 9. Record the presentation using Zoom or any other platform of your choice and upload the recorded video in the assignment portal.

Submission Guidelines:

[1] Submit a single PDF file containing the solutions and interpretations (when needed) to all these problems.
[2] Upload the recorded video in the assignment portal.